

Software Requirements Specification (SRS)

COURSE CODE: CSE 4204 || COURSE TITLE: MOBILE COMPUTING LAB

Semester: Fall 2026

Project Title: NUBTK_COPILOT - AI-Powered University ERP and Career Architect

Instructor: Md. Riaz Mahmud, Assistant Professor, Dept of CSE, NUBTK

Section: 8D | **Team ID:** CSE4204-8D-T01

GitHub Repository: <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>

1. Team Information

Name	Student ID	Role & Responsibility
Bashir Alam	11220320963	Team Leader. Django backend, AI/ML, System Architecture.
Tisha Rani Chakroborty	11220320954	Frontend Developer. Student dashboard and admin pages using HTML and Tailwind CSS.
Fariha Rahman Oishee	11220320967	Database Manager. PostgreSQL tables in Supabase.
Ratul Raha Shuvo	11220320949	Backend Support. Payment gateway integration, PDF generator, and API testing.

2. Introduction

2.1 Project Overview

NUBTK_COPILOT transforms traditional ERP systems from passive academic grade tracking repositories into an active career architecture system. It enables granular tracking of practical technical skills (such as 'Python Loop Optimization' or 'SQL Joins') acquired by students after completing each semester, bridging the gap between standard academic coursework and real-world job market demands.

2.2 Objectives

- To develop a smart ERP system that tracks students' academic records along with a dynamic technical skill inventory module.
- To build an AI Mentor Hub using the OpenRouter API (Claude 3) for real-time programming assistance, career guidance, and gamified skill enhancement.
- To automate crucial university administrative workflows including user registration, payment processing, and weekly analytics reports.

- To calculate a customized 'Career Readiness Score' (0-100) based on cumulative CGPA and market data maps (e.g., BDJobs, LinkedIn) to help students identify their skill gaps.

2.3 Scope

The system consists of a web portal serving three core user roles: Students, Faculty, and Administrators. The architectural implementation encompasses Django REST APIs, a PostgreSQL database layer, an artificial intelligence integration layer, and automated background workers using Celery. Please note that mobile application development is explicitly declared out of scope for the current iteration (Week 3 deliverables).

3. Functional Requirements

Requirement ID	Functional Description
FR-01: User Authentication	Provides comprehensive user registration, login, and logout capabilities with distinct role-based access controls (Student, Faculty, Admin, Super Admin).
FR-02: Skill Inventory Module	Allows students to systematically log technical skills acquired during a course along with valid verification/evidence links for subsequent evaluations.
FR-03: AI Mentor Hub	Integrates interactive chat assistance via OpenRouter API (Claude 3) providing robust coding solutions, career path suggestions, and gamified points.
FR-04: Admin Insight Engine	Deploys automated background tasks via Celery to conduct weekly predictive student data analysis using Scikit-learn and triggers proactive email alerts.
FR-05: Career Readiness Score	Computes a normalized dynamic readiness score (weighted 60% CGPA and 40% technical skills) and maps profiles against active job markets like BDJobs.
FR-06: Payment & Admission	Handles end-to-end automation for registration workflows, payment gateway integrations, and instant automated generation of digital PDF receipts.
FR-07: Academic Dashboard	Provides faculty with an analytical interface to monitor student performance, inspect generalized skill gaps, and download reports.

4. Non-Functional Requirements

- **Performance:** System APIs must guarantee a response time of less than 500ms under standard load conditions. The AI Mentor Hub interface must stream the first response token within 2 seconds of prompt submission.
- **Security:** User passwords must be encrypted using secure PBKDF2 hashing algorithms. All data transmission must follow HTTPS protocol guidelines. Row-Level Security (RLS) configurations must be strictly enforced on Supabase tables.
- **Usability:** The user interface built on Tailwind CSS must be fully adaptive and responsive, supporting standard mobile browser breakpoints down to a minimum width of 400px.
- **Scalability:** The application infrastructure must be ready for seamless deployment on cloud hosting services like Render or Railway, scaling efficiently to manage transactional loads for more than 10,000 active students.
- **Availability & Reliability:** The core web portal must ensure an operational uptime of 99.5%. Automated background tasks handled via Celery should execute a maximum of 3 retries in the event of worker or network failures.

5. User Roles & Access Control

- **Student Role:** Authorized to view academic records, log practical skill entries with validated URLs, chat with the AI Mentor Hub, and view their personalized Career Readiness Score and industry gap feedback.
- **Faculty / Admin Role:** Granted permissions to monitor overall student academic records, analyze aggregated engineering skill graphs, and receive proactive automated system alerts regarding at-risk students.
- **Super Admin Role:** Maintains unconditional global access to manage user authentications, institutional courses, system configurations, and automated digital financial payments.

6. Use Cases

- **UC-01: Log Skill** — Actor: Student. The student selects a completed course from their dashboard, inputs a newly acquired granular skill with documentation evidence, and the backend system validates and appends it to the database records.
- **UC-02: Ask AI Mentor** — Actor: Student. The student types an engineering or career query into the AI Mentor interface. The request handles data exchange through the OpenRouter API to stream prompt responses via Claude 3 while distributing gamified points.
- **UC-03: Weekly Alert Automation** — Actor: Celery System Worker. On an automated weekly Sunday schedule, the Celery background worker invokes predictive machine learning scripts using Scikit-learn to parse student performance and automatically routes high-priority email warnings to faculty admins regarding at-risk trajectories.
- **UC-04: View Career Readiness Score** — Actor: Student. Upon accessing the main dashboard, the application dynamically queries the student's active CGPA and registered skills, computes the composite readiness score, and builds a comprehensive visual map indicating current workforce job market gaps.

7. System Diagrams

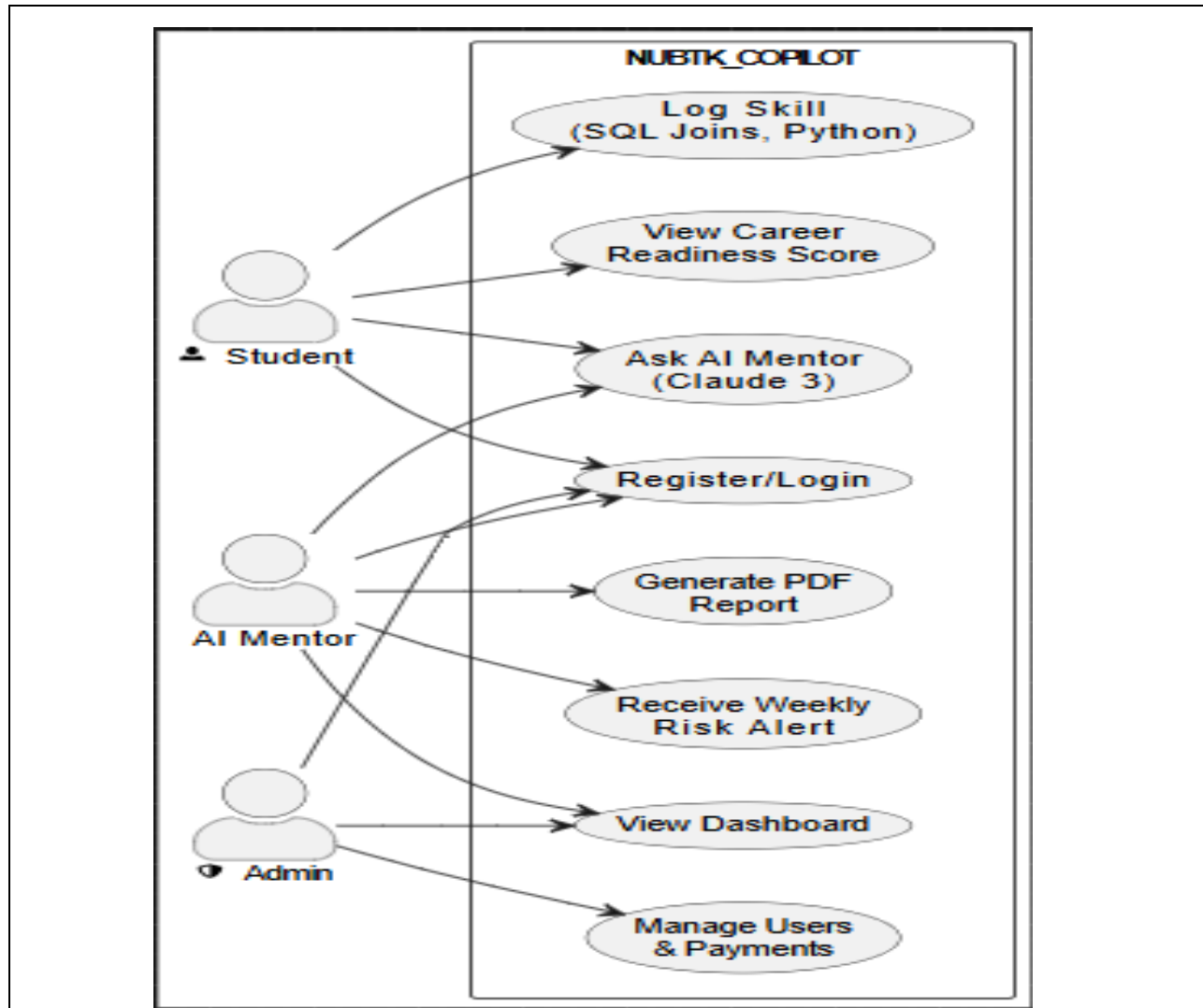


Figure 1: System Use Case Diagram Layout

8. ER Diagram

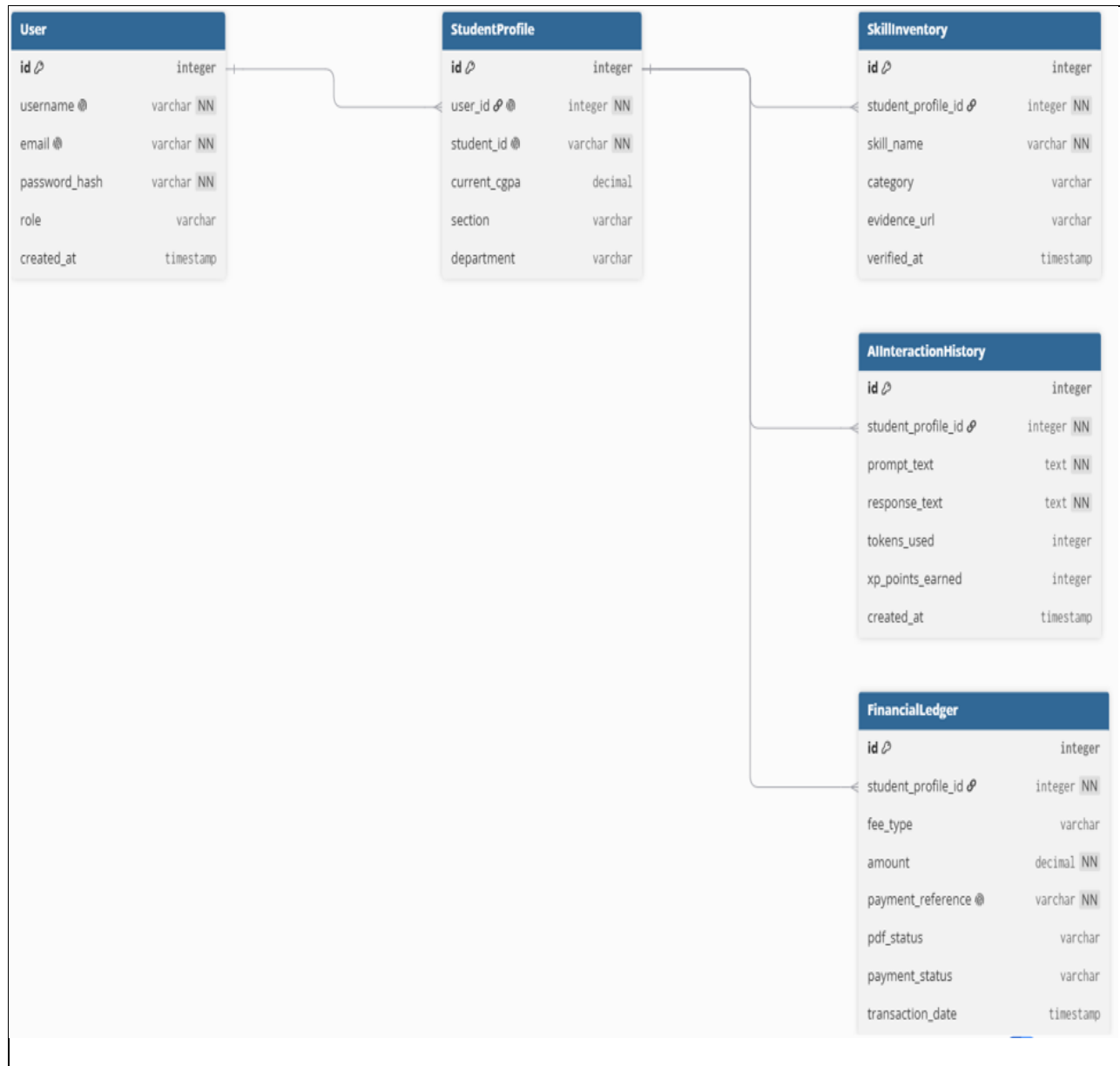


Figure 2: ER Diagram